

Report - Effectiveness of isolation and contact tracing to control Bundibugyo virus (Ebola disease) outbreak in the Democratic Republic of the Congo

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Introduction

As of 24 May 2026 there were 906 cumulative suspected cases and 105 confirmed cases of Ebola virus disease caused by the Bundibugyo virus in the ongoing epidemic¹. The outbreak is primarily in the eastern region of the Democratic Republic of the Congo (DRC), mostly in the Ituri Province, but with confirmed cases in North Kivu and South Kivu¹. There are seven confirmed cases in Uganda, three imported from the DRC and others secondary cases among their contacts². On 17 May 2026 the World Health Organization (WHO) declared the Bundibugyo virus outbreak a public health emergency of international concern (PHEIC)³.

There is likely an underascertainment of the total cases involved in the Bundibugyo virus outbreak. A report from Imperial College London estimated the total outbreak size on 20 May 2026 was likely 400–900 cases, with a possibility of more than 1,000 cases, when the number of suspected cases reported was 516⁴. A second analysis of the true outbreak size used a joint modelling approach found the most likely total number of Ebola cases on 23 May 2026 to be 930 (90% CrI 483–2234)⁵.

Genetic evidence points to a single spillover event with an outbreak start date around 11 April 2026⁶.

In short, there exists strong evidence for a growing epidemic ($R_t > 1$)¹, which will require an active response to control. Bundibugyo ebolavirus has no licensed vaccine or therapeutic, unlike Zaire ebolavirus, for which both a vaccine (Ervebo) and monoclonal antibody therapies (Inmazeb and Ebanga) have been approved⁷. Therefore, Public Health and Social Measures (PHSMs), a type of non-pharmaceutical intervention, are likely the response options in the short term. Standard PHSMs include the isolation of suspected or confirmed (test positive) Ebola cases, and tracing of their contacts⁸.

We assess options to control the ongoing Bundibugyo virus outbreak in the DRC using a stochastic individual-level branching-process model. We calibrate the model to current, and highly uncertain,

surveillance data by conditioning simulated trajectories on the observed early outbreak size assuming a single spillover event. We then use the calibrated model to evaluate two responses: the proportion of contacts of known cases that are successfully traced (0%–100%), and the speed at which symptomatic cases are isolated after symptom onset (mean 1, 3 or 5 days). We simulate the introduction of various combinations of these components mid-outbreak, assuming PHSMs are activated at the declaration of the public health emergency, across plausible values of the basic reproduction number (R_0), to estimate the probability of bringing the outbreak under control within 3, 6 and 12 months of activating the response.

Methods

We use a stochastic individual-level branching process model of disease spread in the population. We assume a single index case initiates the outbreak and that transmission in the community is defined by a Negative Binomial offspring distribution with a mean equal to the basic reproduction number (R_0), and a dispersion parameter (k) to define the individual-level variability in transmission. At present there are no precise estimates for the R_0 for the current Bundibugyo outbreak. We set R_0 to 2.0 and 3.0, based on recent growth in suspected and confirmed cases¹, to explore the sensitivity of our results to assumed Ebola transmissibility. There is little direct evidence to inform the individual-level heterogeneity in transmission for Bundibugyo virus, however previous Zaire outbreaks have yielded estimates. A systematic review of Ebola virus disease found that most estimates show moderate to high heterogeneity ($k < 1$)⁹. We use the estimate of Polonsky et al.¹⁰ from the 2018–2020 Ebola epidemic in the DRC ($k = 0.27$).

Ebola cases are isolated after a delay from symptom onset. We model this delay from onset-to-isolation as a Gamma distribution. To evaluate how the speed of isolating cases with a positive test result impacts the ability to bring the outbreak under control, we run scenarios with a mean onset-to-isolation delay of 5 days, 3 days and 1 day (all assuming a standard deviation of 1 day). Cases can be isolated earlier than the delay after symptom onset if they are traced as a contact of a known case (i.e. their infector). When a contact is traced they are isolated as soon as their infector is isolated (or after their delay from symptom onset if earlier). Isolation is assumed to reduce all further transmission ($R_0^{\text{iso}} = 0$). We assess how contact tracing ascertainment influences the probability of bringing the outbreak under control, varying the percentage of contacts traced between 0% and 100% at 20% intervals.

To simulate the speed of transmission the model samples the generation time by first sampling each case's symptom onset time from the incubation period, then drawing the infection time of each secondary case from a skew-normal distribution centred on the infector's symptom onset and parameterised by the proportion of transmission that occurs before symptom onset¹¹. We use an incubation period estimated from the 2014 Ebola outbreak in West Africa (Gamma distribution, mean = 10.3 days, SD = 8.2 days)¹². This is similar to the reported incubation period of the Bundibugyo virus (2–21 days,⁷). We assume that no Bundibugyo virus transmission events occur prior to symptom onset (0% presymptomatic transmission), which is consistent with previous Ebola outbreaks¹³. Our epidemic model assumes a symptom-based isolation and contact tracing system, however we assume all infectious Bundibugyo cases are symptomatic (0% asymptomatic) again relying on evidence from past Ebola epidemics¹⁴.

To simulate scenarios consistent with the ongoing Bundibugyo virus outbreak in the DRC, we conditioned each simulation (single index case) on having a cumulative case count in [440, 2230] in at least one of weeks 5, 6 or 7 of the outbreak. Conditioning was implemented by rejection sampling: branching-process simulations were re-run with independent random seeds until each replicate produced a trajectory meeting the criterion, with up to 1,000 attempts permitted per accepted replicate.

We retained 200 accepted replicates per scenario. We assume that no PHSMs (i.e. isolation of cases or contact tracing) are active for the first 42 days of the outbreak (from the spillover infection). After day 42 we assume that individuals that are symptomatic or contact traced from a symptomatic individual are tested and isolated if they receive a positive test result to prevent further infections. We simulate the outbreak for a maximum of 2 years (730 days).

We assess the impact of only some untraced contacts passively proceeding to isolate (e.g. showing up to Ebola Treatment Units of their own accord). In a response that relies on confirmatory testing or presumptive diagnosis, imperfect test sensitivity and/or ambiguous symptoms means that a proportion of true Bundibugyo cases are not recognized and do not isolate. We model both situations using the same model parameter, *diagnostic sensitivity*, and vary it across 50%, 75% and 100%. Reduced sensitivity against Bundibugyo virus is plausible: there are no diagnostic assays specifically developed or validated for Bundibugyo, and the rapid antigen tests and most widely deployed PCR-based assays were developed and validated primarily against Zaire ebolavirus¹⁵. Likewise, patients with Bundibugyo infections may be presenting differently from patients in other recent ebolavirus outbreaks. Symptomatic cases that are not isolated due to misdiagnosis continue to transmit in the community; cases isolated after symptom onset via contact-tracing follow-up are assumed to always successfully isolate.

We define outbreak control in terms of extinction within the branching process model, and report the date of control as the last case onset plus 42 days, consistent with criteria to declare the end of an ebolavirus outbreak. In our simulation setup we assume that if the cumulative number of infections exceeds 10,000 then that specific outbreak is considered uncontrollable. We test for outbreak control (extinction) at 3, 6 and 12 months from the activation of the PHSMs.

Results

The R_0 for transmission in the community impacts the ability to control this Bundibugyo Ebola epidemic (Figure 1). There is a remote chance ($< 5\%$) of ending the outbreak within 3 months from the activation of PHSMs, namely isolation of cases and contact tracing. Only in the scenario where $R_0 = 2$ and 100% of contacts for infectious individuals are traced does the outbreak have a highly unlikely ($> 15\%$) chance of ending (extinction) within 3 months with NPIs (Figure 1).

If R_0 is around or below 2 then contact tracing at least 80% of contacts shows a realistic to highly likely possibility of outbreak control within 6–12 months of PHSMs being active (Figure 1). If community R_0 is 3 then 100% of contacts need to be traced to have more than a remote chance of control, and even with 12 months of interventions, there is less than a highly likely chance of control (Figure 1).

Proportion of BVD outbreaks controlled

Proportion of outbreaks controlled within 3, 6 or 12 months from the start of PHSMs (42 days after first case)

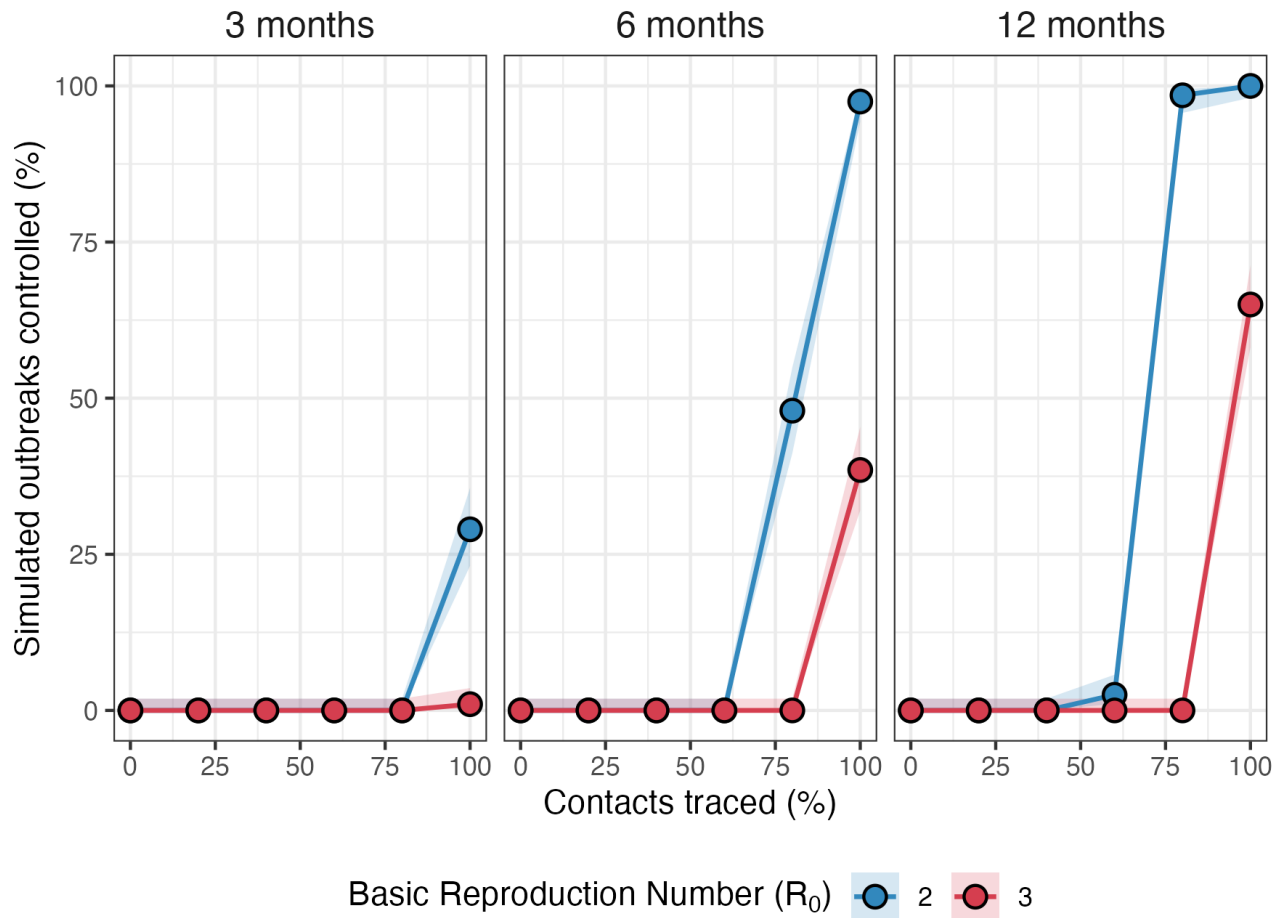


Figure 1: Proportion of simulated Bundibugyo virus outbreaks controlled (no new cases by the evaluation horizon) within 3, 6 and 12 months of PHSM activation, as a function of the percentage of infectious contacts traced (0%, 20%, 40%, 60%, 80%, 100%). Each point is the proportion of 200 conditioned simulations meeting the control criterion; shaded ribbons are 95% Wilson binomial confidence intervals around each proportion. PHSM activation is modelled as day 42 of the outbreak. The mean onset-to-isolation delay is fixed at 3 days (Gamma, SD = 1 day). Colours distinguish basic reproduction numbers $R_0 = 2.0$ and 3.0 .

All results described above assume a mean time delay of 3 days from symptom onset to isolation. Assuming a longer mean delay to isolating cases of 5 days does not drastically alter the results of the 3-day delay, with probabilities of control marginally worse (for all scenarios not at 0% or 100% of outbreaks controlled) (Figure 2). If cases can be isolated on average 1 day after symptom onset and $R_0 \sim 2$, outbreak control is likely within a year if at least 20% of contacts are successfully traced. Outbreak control within 12 months is highly likely if contact tracing can ascertain $> 40\%$ of contacts. When $R_0 > 2$ contact tracing needs to be comprehensive (close to 100%) for probable control within a year, and even then, the chance of control within 6 months is at best only a realistic possibility (Figure 2).

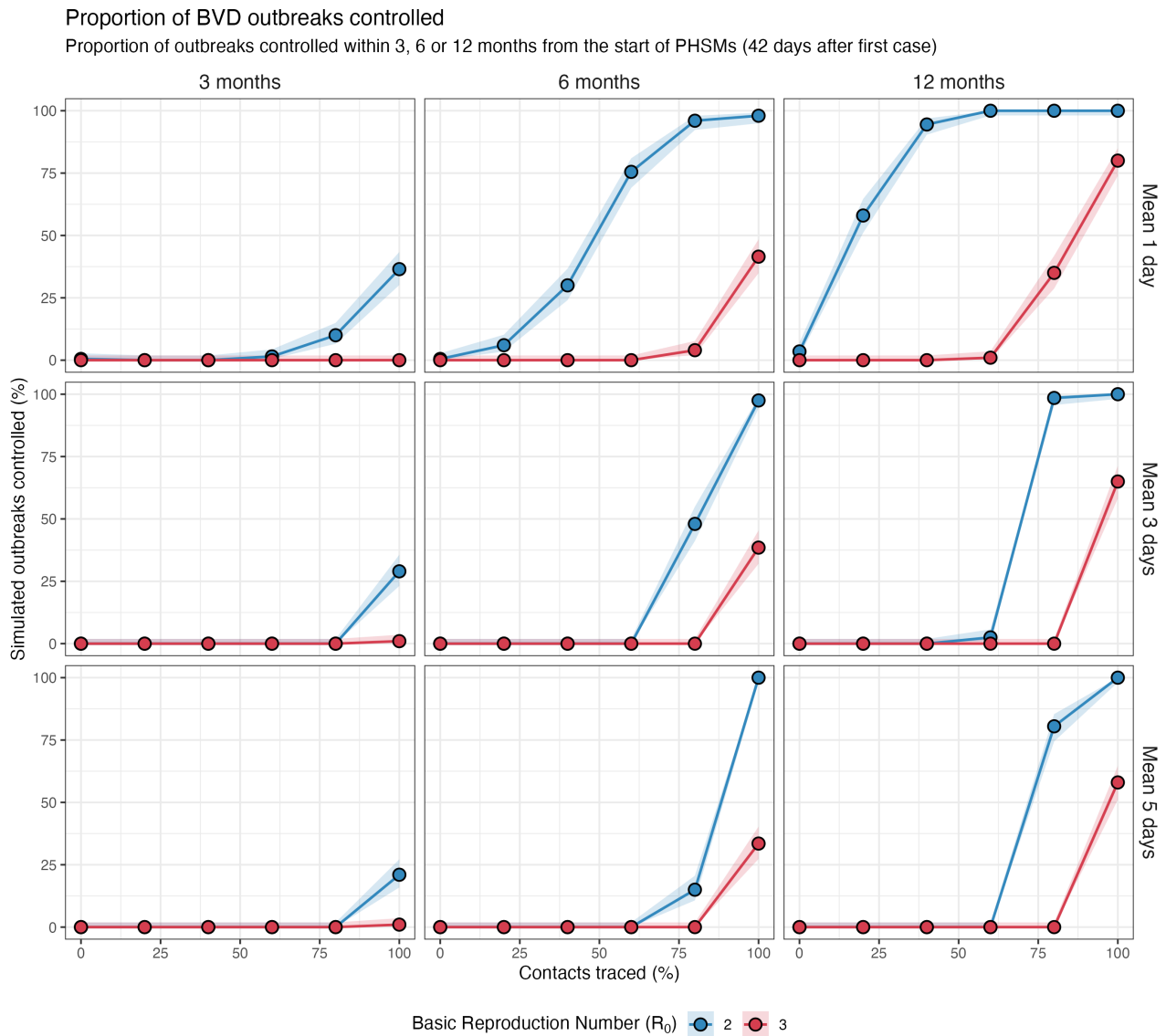


Figure 2: Proportion of simulated Bundibugyo virus outbreaks controlled (no new cases by the evaluation horizon) within 3, 6 and 12 months (columns) of PHSM activation, as a function of the percentage of infectious contacts traced. Rows show three mean onset-to-isolation delays (top to bottom: 1, 3 and 5 days; Gamma, SD = 1 day). Each point is the proportion of 200 conditioned simulations meeting the control criterion; shaded ribbons are 95% Wilson binomial confidence intervals around each proportion. PHSM activation is modelled as day 42 of the outbreak. Colours distinguish basic reproduction numbers $R_0 = 2.0$ and 3.0 .

The sensitivity to correctly diagnose Ebola virus disease cases, either from a positive test or a suspected/probable case based on symptoms, can have a substantial impact on the ability for PHSMs to contain the outbreak (Figure 3). Notably, in a scenario that all community cases are correctly identified, the chance of outbreak control within 3 months, assuming a 1 day average delay from symptom onset to isolation, are greatly improved (Figure 3 topleft panel), and there is an almost certain chance of containment within 12 months at any contact tracing coverage if $R_0 \sim 2$ (Figure 3 topright panel). Across all symptom onset to isolation delays and R_0 values modelled, when diagnostic sensitivity is 50% then the chance of outbreak control with 12 months is remote (Figure 3).

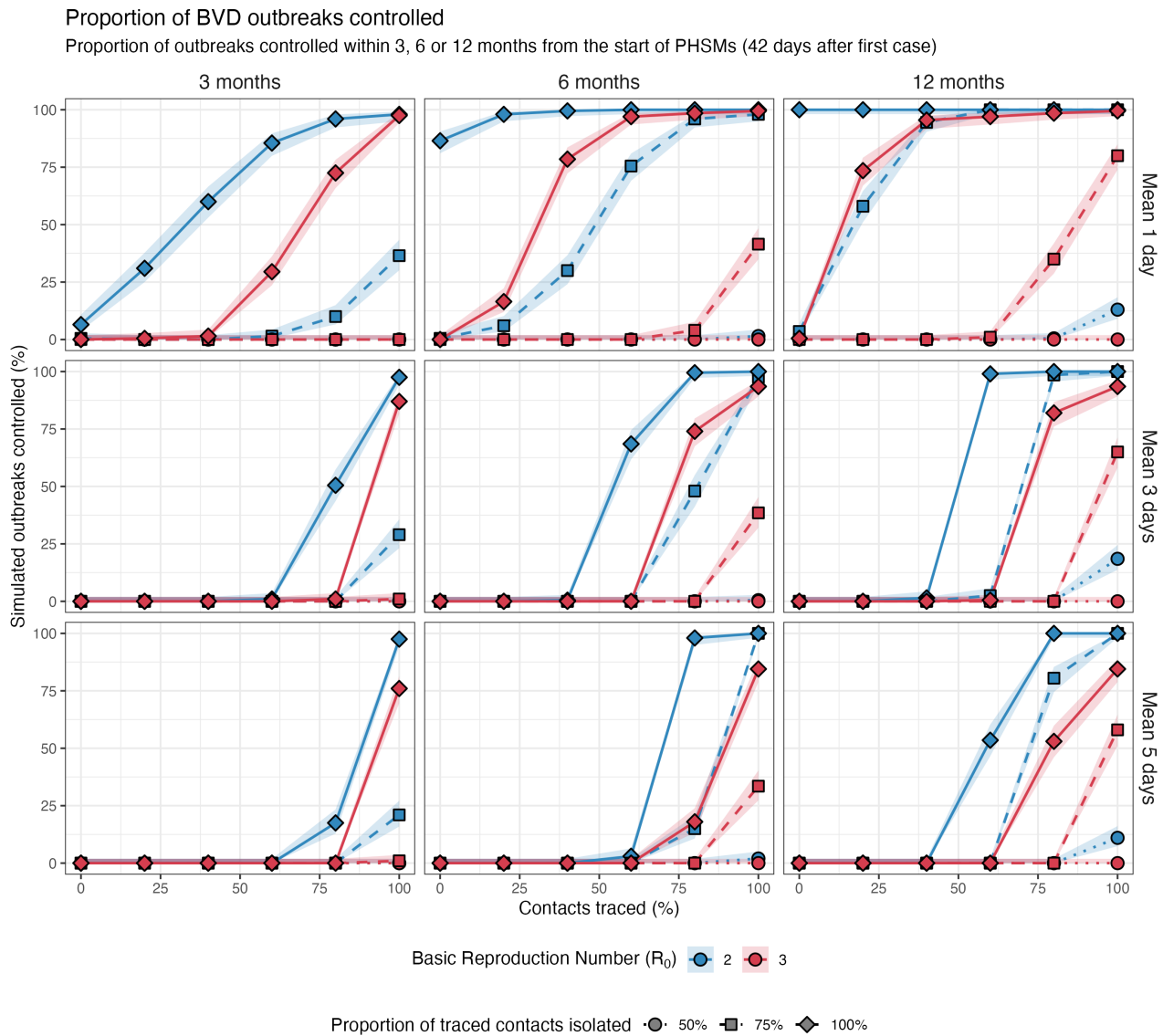


Figure 3: Proportion of simulated Bundibugyo virus outbreaks controlled (no new cases by the evaluation horizon) within 3, 6 and 12 months (columns) of PHSM activation, as a function of the percentage of infectious contacts traced. Rows show three mean onset-to-isolation delays (top to bottom: 1, 3 and 5 days; Gamma, SD = 1 day). Point shape and line style indicate three values of the proportion of untraced infections isolated: 50% (circle, dotted), 75% (square, dashed) and 100% (diamond, solid). Each point is the proportion of 200 conditioned simulations meeting the control criterion; shaded ribbons are 95% Wilson binomial confidence intervals around each proportion. PHSM activation is modelled as day 42 of the outbreak. Colours distinguish basic reproduction numbers $R_0 = 2.0$ and 3.0 .

The descriptions of probability in this report correspond to those in the UK Government Probability Yardstick¹⁶.

Discussion

Overview

In this report we have assessed the effectiveness of targeted PHSMs by identifying and isolating individuals with Ebola virus disease and tracing and isolating their contacts. We have evaluated how contact tracing coverage and speed of isolating symptomatic individuals impacts the probability of

containment. By varying the speed of isolation we hope to cover responses that isolate suspected or probable cases, or those that require a positive test result⁸. We find that controlling the outbreak within 3 months is unlikely. Only with a fast (mean isolation <3 days from symptom onset to isolation) and comprehensive ($>80\%$ contacts traced) response do we find probable outbreak control within a year of interventions being active. Assuming community cases are always correctly identified shortens the time to elimination, with a realistic possibility of the outbreak lasting less than 6 months.

By incorporating the delay to activation of non-pharmaceutical measures we hope to more accurately estimate what control measures are effective and feasible for this outbreak, rather than assuming the intervention is activated by the index case, which would be the case if the outbreak were known and imported into a different region¹¹.

By evaluating outbreak control (extinction) at 3, 6 and 12 months from the activation of PHSMs, we provide a rough indication of the potential duration of the Bundibugyo outbreak. The majority of our simulated scenarios have a $<50\%$ chance of outbreak control within 6 months. Diagnostic sensitivity below approximately $1 - 1/R_0$ ($\approx 50\%$ at $R_0 = 2$, $\approx 67\%$ at $R_0 = 3$) allows a self-sustaining chain of undiagnosed cases that contact tracing alone cannot interrupt, since traced contacts of an unisolated infector also remain outside the response in our model¹⁷. Together these results indicate that rapid development of pharmaceutical interventions, such as vaccines, may be required to help eliminate Bundibugyo transmission within the coming months¹⁸.

Limitations

Our model does not account for the geographic spread of the Bundibugyo virus, and any geographical variation in response speed or capacity. Our analysis aims at informing what targeted PHSMs would be required to bring the outbreak to an end, and what outbreak duration is likely under different epidemiological characteristics until more precise estimates of parameters such as the basic reproduction number or the incubation period for Bundibugyo become available. There are reports of cases infected in healthcare facilities ($\sim 50\%$) where contact tracing is less effective, and our model does not incorporate these setting-specific factors.

Given the limited estimation of epidemiological characteristics of the Bundibugyo virus in this outbreak we have based the incubation period and individual-level heterogeneity (k) on past estimates from the literature. Information on the transmission network would allow a better estimate of the individual-level heterogeneity in transmission¹⁹. If transmission is more homogeneous between people in the community then, theoretically, the outbreak would be less controllable²⁰. We assess how sensitive our results are to changes in pathogen transmissibility (R_0), onset-to-isolation delays, and diagnostic sensitivity; however, these too are not precisely calibrated to early outbreak data¹, ongoing emergency response efforts, or reported test performance respectively. Additionally, in future work we could model the improved test sensitivity as new diagnostics specific to Bundibugyo become available.

We do not model capacity constraints on testing, isolating and contact tracing. These are resource-limited countermeasures, and if the number of suspected cases continues to rise these may become overwhelmed, reducing the efficacy of the intervention¹⁷. On 21 May 2026 the WHO reported contact tracing follow-up rate is 21% ²¹.

Our model assumes that traced contacts are followed up and correctly diagnosed. An alternative scenario would assume that diagnostic sensitivity applies equally to traced and untraced symptomatic

infections; under that assumption the efficacy of contact tracing would be lower, and the structural sensitivity threshold described above would shift further from feasibility.

Conclusion

In conclusion, we hope that this report provides evidence to inform an effective control strategy for the ongoing epidemic caused by Bundibugyo ebolavirus using a mathematical modelling approach, as has been conducted for previous Ebola outbreaks²². Operationally, our findings suggest that the response should prioritise: (i) achieving high contact-tracing coverage as soon as practicable after activation; (ii) deploying diagnostic capacity with sensitivity adequate for Bundibugyo virus; and (iii) planning logistics for an outbreak that may persist for many months even under a well-resourced response.

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Code availability

The code to reproduce this analysis can be found at: <https://github.com/joshwlambert/BundibugyoNPI2026>

The outbreak model code can be found in the R package `ringbp` at <https://github.com/joshwlambert/ringbp@develop>. This analysis used the `develop` branch of the `ringbp` package which can be installed with:

```
remotes::install_github("joshwlambert/ringbp@develop")
```

AI Disclaimer

This work was assisted by the use of Claude Opus 4.7.

Supplementary Material

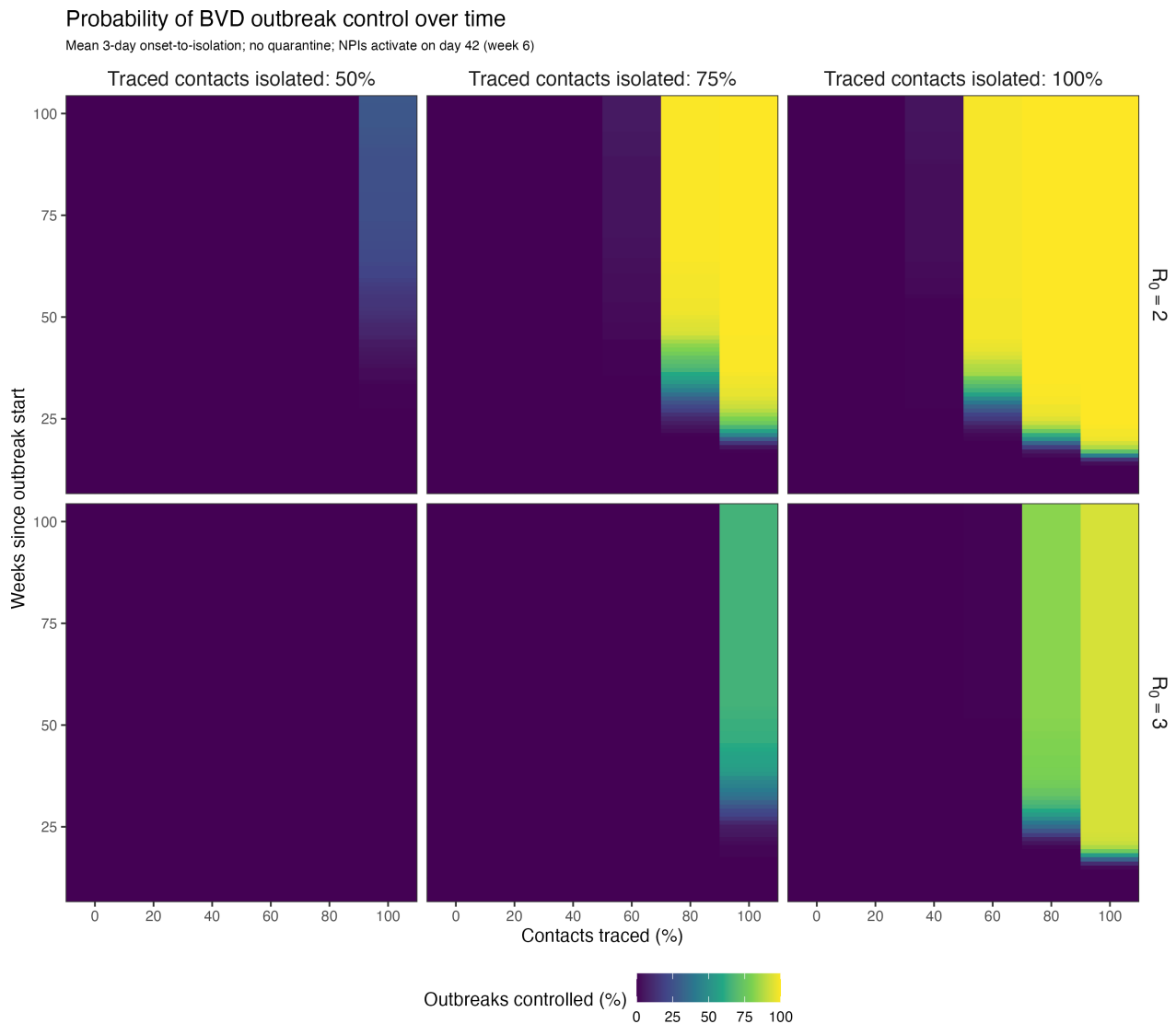


Figure 4: Probability of Bundibugyo virus disease outbreak control as a function of the percentage of contacts traced (x-axis) and the time elapsed since the outbreak start (y-axis, weeks). Rows correspond to the assumed basic reproduction number ($R_0 = 2$ and $R_0 = 3$); columns correspond to the assumed test sensitivity (50%, 75% and 100%). The fill colour shows the proportion of conditioned simulations that had reached extinction by the given week. Restricted to the mean 3-day onset-to-isolation scenario. PHSMs activate on day 42 of the outbreak (week 6).